Adaptive Temperature-Driven Ternary Contrastive Autoencoder Framework for Hyperspectral Target Detection

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***Abstract*—Hyperspectral target detection (HTD) is a critical task in remote sensing, where numerous deep learning (DL)-based methods have emerged for their powerful ability to extract hierarchical and discriminative features. However, challenges such as insufficient labeled samples and spectral variability lead to formidable issues for DL-based methods like model underfitting and poor robustness. In particular, existing contrastive learning (CL)-based detectors rely on native positive–negative pair construction while overlooking hidden positive pairs (i.e., positive but mistakenly constructed as negative), which undermines the model's ability to maintain consistent feature representation. To address the above issues, we propose an adaptive temperature-driven ternary contrastive autoencoder (ATTCA) framework, which performs HTD in a self-supervised manner. Initially, we introduce a novel augmentation technique named strong–weak frequency-domain interference (S-WFI) to expand data while establishing the ternary framework, which can balance the robustness and representation consistency of the model. Additionally, a dual-stream quad-scan Mamba (DSQM) network based on a compositing selective state-space model (SSM) is tailored to effectively extract multiscale spatial–spectral features and mitigate the effects of spectral variation. Ultimately, we formulate a reconstruction weight-driven adaptive temperature (RWAT) strategy to dynamically adjust parameters and suppress the separation of hidden positive pairs, which can facilitate the alignment of target features effectively. Experimental results on four real-world benchmark datasets demonstrate that our approach outperforms state-of-the-art methods in terms of detection performance and efficiency.**

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I. Introduction

HYPERSPECTRAL images (HSIs) are 3-D data cubes with two spatial dimensions representing pixel coordinates and one spectral dimension measuring reflectance across bands [1], [2]. Due to their unique characteristics, HSIs have been extensively applied in various practical fields, such as Earth observation [3], environmental monitoring [4], and mineral exploration [5]. Hyperspectral target detection (HTD), a significant task in HSIs processing, aims to locate targets through estimating the likelihood of the known spectral signature within each pixel [6]. Unlike bounding-box-based object detection, which annotates spatial location using rectangular boundaries, HTD localizes targets and separates them from backgrounds at the pixel or subpixel level [7], [8].

Over the past few decades, numerous target detection methods have been proposed. Classical spectral matching models, such as the Euclidean distance and spectral information divergence, are inspired by similarity measures, which distinguish targets through spectral difference. Furthermore, abundant statistics and assumption-based models are extensively utilized, including matched filter (MF), constrained energy minimization (CEM), and orthogonal subspace projection (OSP) [10]. However, the aforementioned models are based on prior assumptions, such as Gaussian distributions, and often struggle to achieve satisfactory results in real-world applications. In contrast, some representation-based models [11], [12] operate without relying on assumptions, which formulate pixels as a linear combination of atoms and detect targets through reconstruction errors. Nevertheless, they rely on manually selected intervention and low-order statistics, making it difficult to effectively capture the nonlinear characteristics of HSIs. Then, some detectors incorporate the kernel concept [13] or the bilinear model [14] to compensate for nonlinearity, partially resolving the challenge.

Due to its powerful discriminative nonlinear feature extraction capability, deep learning (DL) has been widely employed in HTD. However, the intrinsic complexity of HSIs, such as the curse of dimensionality and spectral variability, imposes significant challenges on DL-based methods, including the

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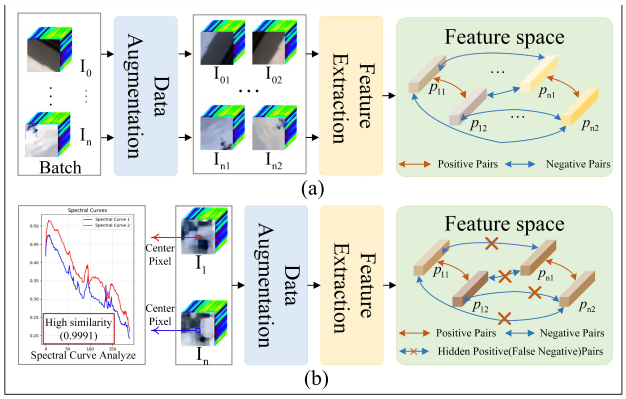


Fig. 1. (a) General CL framework (e.g., SimCLR [9]) and its positive and negative pairs construction strategy (two augmented samples p_{11} and p_{12} from the same instance p_1 form a positive pair, while two augmented samples p_{11} and p_{n1} from different instances p_1 and p_n form a negative pair). (b) Hidden positive pairs phenomenon in the general framework (augmented samples from high semantic similarity instances are not true negative pairs, but rather HP).

scarcity of labeled samples and intraclass spectral inconsistency. Nowadays, numerous algorithms based on convolutional neural network (CNN) [15], [16], Transformer [16], [17], and state-space models (SSMs) [18] have achieved critical breakthroughs in addressing the aforementioned challenges and can be broadly categorized into two types.

The first category follows a supervised learning paradigm, where sufficient samples are augmented from prior spectral data to effectively capture spectral variability and train robust binary classifiers. For instance, Zhang et al. [19] combined autoencoder and linear prediction strategies to expand training samples. Subsequently, Gao et al. [20] innovated a target random mask technique to augment samples for training the network. Although these methods employ remarkable augmentation techniques, some samples may still deviate from the true distribution (i.e., overaugmentation samples). This deviation introduces noise into the detectors, which ultimately degrades the detection accuracy.

Another approach designs auxiliary tasks to extract supervisory information from unlabeled data for self-supervised training [21]. In particular, contrastive learning (CL) is an exemplary architecture. As illustrated in Fig. 1(a), general CL approaches construct positive and negative pairs via tailored augmentations and apply CL loss to bring positives closer and separate negatives, thus learning effective features. Constructing diverse positive pairs without manual labels, CL effectively alleviates the curse of dimensionality and enables the model to learn semantically consistent features against spectral variability. For example, Wang et al. [22] employed odd and even band selection operations for data augmentation, achieving HTD through CL at the spectral level. Shen et al. [18] introduced a CL-based HTD framework incorporating a pyramid SSM, which leverages spectral contrast to effectively differentiate targets from the background.

The aforementioned algorithms primarily focus on designing data augmentation techniques and building novel feature extraction networks but overlook certain inherent issues within the general framework.

1) Existing CL-based detectors rely on inherent sample pair construction strategies, overlooking the noise introduced by overaugmentation, thus failing to effectively balance model robustness and feature representation consistency.
2) Existing CL-based detectors overlook hidden positive pairs (HP) in Fig. 1(b), thereby erroneously repelling semantically correlated target samples within the embedding space, which induces target feature dispersion and compromises detection performance in HTD.

To address the aforementioned challenges, an adaptive temperature-driven ternary contrastive autoencoder framework (ATTCA) for HTD is proposed. First, a customized strong–weak frequency-domain interference (S-WFI) technique is proposed for HSIs to augment samples via graded phase and amplitude interference. An original sample forms two positive pairs with its own strong/weak augmented samples while constructing negative pairs with augmented samples from other instances, thereby establishing a novel ternary CL framework. Introducing weak augmentation with constrained frequency-domain interference, S-WFI suppresses the distributional drift of overaugmentation while maintaining semantic consistency, thus balancing model robustness and feature representation consistency. Furthermore, a dual-stream quad-scan Mamba (DSQM) network based on compositing SSMs (CS6) is conceived to mitigate the effects of spectral variability. Unlike existing SSM-based networks, DSQM bridges pixel and patch streams to extract spatial–spectral features for stronger discriminability. Meanwhile, DSQM performs parallel scanning of the feature sequence in four directions to capture long-range dependencies, which can extract multiscale features with linear computational complexity and fewer parameters. In addition, we propose a reconstruction weight-driven adaptive temperature strategy (RWAT), which computes weights from the reconstruction errors of targets and backgrounds to dynamically adjust the temperature of HP. Compared to a fixed temperature, RWAT adaptively regulates the emphasis on HP during gradient backpropagation. This promotes better feature alignment for target samples, thereby reducing intraclass variance and enhancing interclass discrimination. The main contributions of our work can be outlined as follows.

1) In this article, we propose an innovative self-supervised ternary CL framework for HTD.
2) To balance the robustness and consistency of the model, we develop a tailored S-WFI technique for HSIs to augment the data and construct a ternary contrastive framework.
3) A streamlined DSQM network is proposed to capture multidirectional long-range dependencies with linear complexity, extracting multiscale spatial–spectral features to mitigate spectral variation.
4) To align target features, we implement the RWAT strategy to dynamically adjust the temperature of hidden positive pairs, suppressing their dispersion and enhancing intraclass similarity and interclass separability.

The rest of this article can be summarized as follows. Section II presents the relevant knowledge covered in this article. Section III describes our proposed method in detail.

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Section IV presents the experimental results and analysis. Section V concludes this article.

II. Related Work

A. Self-Supervised CL (SSCL)

SSCL [23] has become a powerful paradigm for HSI representation learning, enabling effective feature extraction without manual annotations. SimCLR [9], a representative SSCL framework, constructs positive pairs via stochastic augmentations (e.g., cropping and flipping) of the same image while treating other instances in the batch as negatives, as shown in Fig. 1(a). A deep network encodes these augmented samples into latent features, and contrast loss maximizes similarity between views of the same instance while separating different instances. Despite its success, most HSI detectors still rely on a fixed contrastive formulation and overlook the HP issue illustrated in Fig. 1(b), where semantically similar samples may be incorrectly treated as negatives, limiting feature discriminability and representation consistency. To overcome this limitation, we propose an innovative self-supervised ternary CL framework that explicitly identifies and optimizes HP, achieving more efficient and balanced CL tailored for HTD.

B. State-Space Model

The SSM [24] is a structured sequential modeling architecture that integrates the advantages of recurring neural networks (RNNs) and CNNs. Unlike Transformers, SSM leverages structured state-space computation with $O(\text{NlogN})$ complexity. Derived from continuous systems, SSM extracts information from the input $u(t) \in \mathbb{R}$ and models the sequence through the hidden state $h(t) \in \mathbb{R}^N$ to ultimately produce a prediction $y(t) \in \mathbb{R}$. Typically, SSM can be represented by the following set of linear differential equations:

$$\begin{aligned} h'(t) &= \mathbf{A}h(t) + \mathbf{B}x(t) \\ y(t) &= \mathbf{C}h(t) \end{aligned} \tag{1}$$

where $\mathbf{A} \in \mathbb{R}^{N\times N}$, $\mathbf{B}$, and $\mathbf{C} \in \mathbb{R}^{N\times 1}$ are system matrices representing state transfer, input-to-state influence, and state-to-output mapping, respectively.

To apply SSM in DL tasks, the zero-order hold (ZOH) technique discretizes the continuous parameters, resulting in

$$\begin{aligned} \overline{\mathbf{A}} &= \exp(\Delta\mathbf{A}) \\ \overline{\mathbf{B}} &= (\Delta\mathbf{A})^{-1}\left(\exp(\Delta\mathbf{A}) - \mathbf{I}\right)\cdot\Delta\mathbf{B} \\ h_t &= \overline{\mathbf{A}}h_{t-1} + \overline{\mathbf{B}}x_t \\ y_t &= \mathbf{C}h_t. \end{aligned} \tag{2}$$

Furthermore, to overcome the limitations of the linear time-invariant assumption in modeling contextual relationships within input sequences, Mamba [25] proposes the selective SSM (S6). Unlike traditional SSMs that rely on static parameterization, S6 dynamically adjusts the projection matrices based on the input, enabling selective attention for each position in the sequence. Specifically, the parameters $\mathbf{B} \in \mathbb{R}^{B\times L\times N}$, $\mathbf{C} \in \mathbb{R}^{B\times L\times N}$, and $\Delta \in \mathbb{R}^{B\times L\times D}$ are computed adaptively from the input sequence $\mathbf{x} \in \mathbb{R}^{B\times L\times D}$ as follows:

$$\begin{aligned} \mathbf{B} &= \text{Linear}_N(\mathbf{x}) \\ \mathbf{C} &= \text{Linear}_N(\mathbf{x}) \\ \Delta &= \text{Broadcast}_D(\text{Linear}_1(\mathbf{x})) \end{aligned} \tag{3}$$

where $\text{Linear}_N(\cdot)$ denotes a learnable linear projection to dimension N and $\text{Broadcast}_D(\cdot)$ expands the projected result to match the hidden size D.

Thanks to Mamba's flexibility in modeling high-dimensional sequence data, it has been adopted for HTD tasks, such as HTD-Mamba [18], which captures multiresolution spectral information via the S6 architecture. Yet, its spatial–spectral interaction [26] remains limited. We therefore propose a dual-stream network that performs parallel spatial and spectral modeling with feature concatenation to enhance joint representation.

III. Proposed Method

The overall structure of ATTCA is shown in Fig. 2, which consists of a training phase and a testing phase. In training, all patches are first augmented by the S-WFI to construct the ternary CL framework. The augmented data then pass through the DSQM network for feature extraction and balanced ternary CL loss computation. In the preprocessing stage, a prior patch is derived via the calculation of the SAM between the prior spectrum and each patch's central pixel, with the CEM algorithm and density-based clustering employed to secure reliable background data. In conjunction with the original patches, they are fed into an autoencoder to compute reconstruction weights, which are used by RWAT to dynamically adjust the contrastive loss temperature. In testing, the DSQM encoder network is frozen, and test samples along with prior patches are input to obtain the final detection results. Details of S-WFI, DSQM, and RWAT are introduced in Section III.

A. Strong–Weak Frequency-Domain Interference

Given that each pixel in the HSIs represents a continuous high-dimensional spectral signal with distinct frequency characteristics [27], frequency-domain interference serves as an effective augmentation. We introduce the S-WFI, which perturbs the amplitude and phase spectra at the pixel level to generate weakly and strongly augmented samples. Specifically, weak augmentation employs the smaller magnitude perturbations to amplitude and phase spectra, thus preserving core semantics with minimal disturbance; in contrast, strong augmentation applies larger magnitude perturbations, modifying finer spectral details while introducing greater randomness. By adjusting augmentation parameters, S-WFI enables controlled spectral interference in the frequency domain.

Given an input patch $\mathbf{X}_o \in \mathbb{R}^{p^2\times B}$, where p^2 and B represent the number of pixels and bands, respectively, we apply the discrete Fourier transform (DFT) to each pixel's spectral vector to obtain the phase spectrum $\mathbf{X}_{op} \in \mathbb{R}^{p^2\times B}$ and the amplitude spectrum $\mathbf{X}_{oa} \in \mathbb{R}^{p^2\times B}$. By introducing varying levels of interference and performing the inverse DFT (IDFT),

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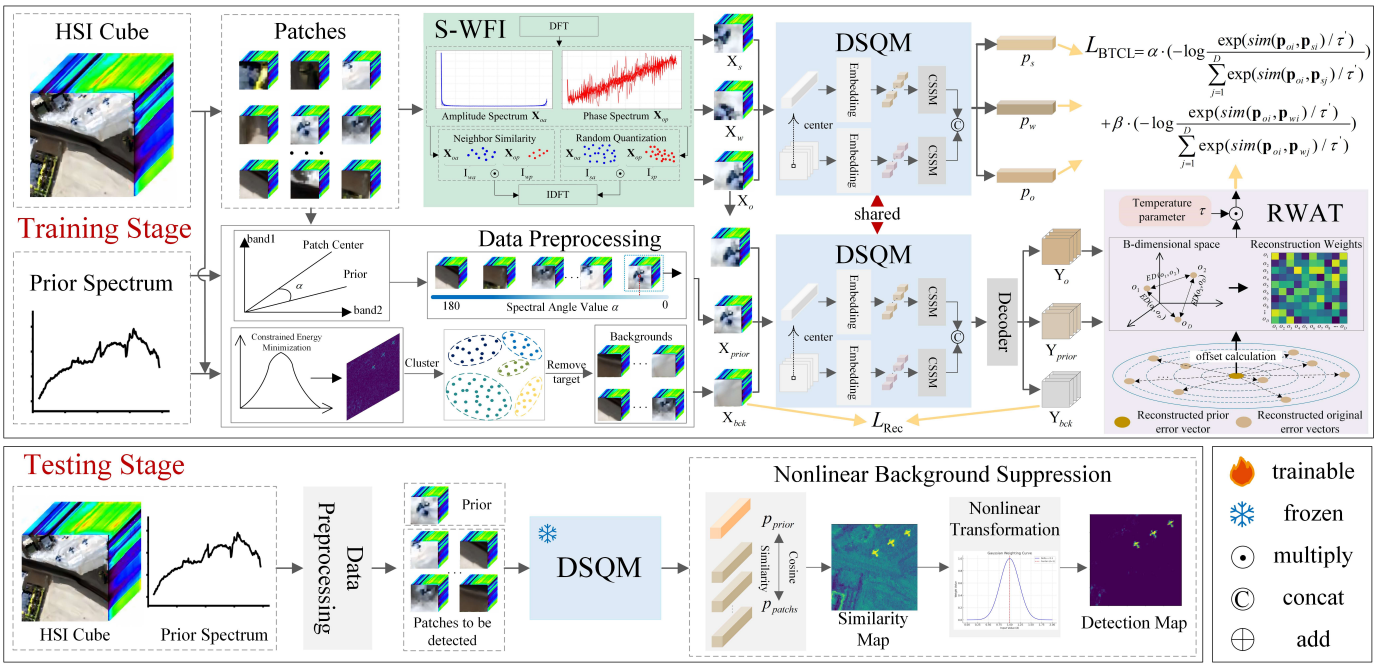


Fig. 2. Overall architecture of the proposed ATTCA. HSI cubes and the prior spectrum are the input of training and testing stage. During training, the feature encoder DSQM is constrained by both a temperature-adaptively adjusted CL loss and an autoencoder reconstruction loss. The trained DSQM remains frozen for feature extraction during testing.

we generate both strongly and weakly augmented samples, as shown in the following:

$$\begin{aligned}\mathbf{X}_s &= \mathrm{IDFT}\left(\mathbf{I}_{sa}\cdot\exp\left(j\cdot\mathbf{I}_{sp}\right)\right)\\ \mathbf{X}_w &= \mathrm{IDFT}\left(\mathbf{I}_{wa}\cdot\exp\left(j\cdot\mathbf{I}_{wp}\right)\right)\end{aligned}\tag{4}$$

where $\mathbf{I}_{sa}$ and $\mathbf{I}_{sp}$ denote the interfered amplitude and phase spectra under strong interference, respectively, and $\mathbf{I}_{wa}$ and $\mathbf{I}_{wp}$ represent those under weak interference.

For strong interference, we employ improved randomized quantization (RQ) [28] on selected spectral channels of both amplitude and phase spectra.RQ maps each value x_i within a quantization bin $\mathcal{S}$ to a randomly sampled reconstruction value y_i, introducing controlled noise. For the given amplitude $\mathbf{X}_{oa}$ and phase $\mathbf{X}_{op}$, we select λ spectral channels and divide their value ranges into N bins. For example, for the kth band $\mathbf{X}_{oa_k}\in\mathbb{R}^{p^2\times 1}$, we randomly sampled the p^2-1 values to define the boundaries of the bin

$$\begin{aligned}&a_0=\min\left(\mathbf{X}_{oa_k}\right),\quad a_{N-1}=\max\left(\mathbf{X}_{oa_k}\right)\\ &a_i\sim U\left(a_0,a_{N-1}\right),\quad i=1,\ldots,N-2\\ &\mathcal{S}=\left\{S_i=\left[a_i,a_{i+1}\right)\right\},\quad i=0,\ldots,N-1.\end{aligned}\tag{5}$$

The quantized reconstruction value y_i is then sampled from the uniform distribution $U(a_i,a_{i+1})$. After processing all selected bands, we combine the perturbed amplitude and phase spectrum and perform the IDFT to reconstruct the strongly augmented sample. By applying the same operations to all pixels and introducing random flipping and rotation, we obtain the final strongly augmented samples $\mathbf{X}_s\in\mathbb{R}^{p^2\times B}$. The number of quantization bins N is treated as a hyperparameter that controls the interference strength.

Weak interference is constructed based on a neighbor correlation metric that quantifies the spectral similarity between the center pixel and its surrounding neighbors. This metric is designed to regulate the augmentation strength by prioritizing spectrally correlated neighbors, thereby maintaining semantic consistency while introducing minimal perturbations. Specifically, for the amplitude and phase spectra $\mathbf{X}_{oa}$ and $\mathbf{X}_{op}$, the center pixel's spectrum is denoted as $\mathbf{P}_{oa}$, $\mathbf{P}_{op}\in\mathbb{R}^{1\times B}$. For the amplitude spectrum, the cosine similarity between the center pixel and its neighbors is calculated to obtain the correlation weights $\mathbf{W}_i$, which are subsequently used to linearly combine the neighboring amplitude spectra

$$\mathbf{I}_{wa}=\sum_{i=0}^{p^2-1}\left(\mathbf{W}_i\cdot\mathbf{X}_{ia}\right),\quad\text{where}\quad\mathbf{W}_i=\frac{\exp\left(S_i\right)}{\sum_{j=0}^{p^2-1}\exp\left(S_j\right)}\tag{6}$$

where S_i is the similarity score defined as

$$S_i=\frac{\mathbf{P}_{oa}^{\mathrm{T}}\mathbf{P}_{oa}^{i}}{\|\mathbf{P}_{oa}\|_2\cdot\left\|\mathbf{P}_{oa}^{i}\right\|_2}\tag{7}$$

where $\|\cdot\|_2$ denotes the $L2$ norm and $\mathbf{P}_{oa}^{i}$ is the amplitude spectrum of the ith pixel. The same approach is used for the phase spectrum. Finally, IDFT is applied to the combined spectra to generate weakly augmented samples, followed by random rotations and flips to produce the final weak augmentation $\mathbf{X}_w\in\mathbb{R}^{p^2\times B}$.

Finally, we generate strongly and weakly augmented samples to modify fine-grained spectral details and preserve core semantics, respectively. Each original sample is paired with these two augmented versions to form positive pairs, while other augmented samples serve as negatives, resulting in a balanced ternary CL framework. By jointly training both branches, the framework dynamically balances robustness and consistency in feature representation. Experiments in Section IV show that a 1:1 ratio of strong to weak augmentations

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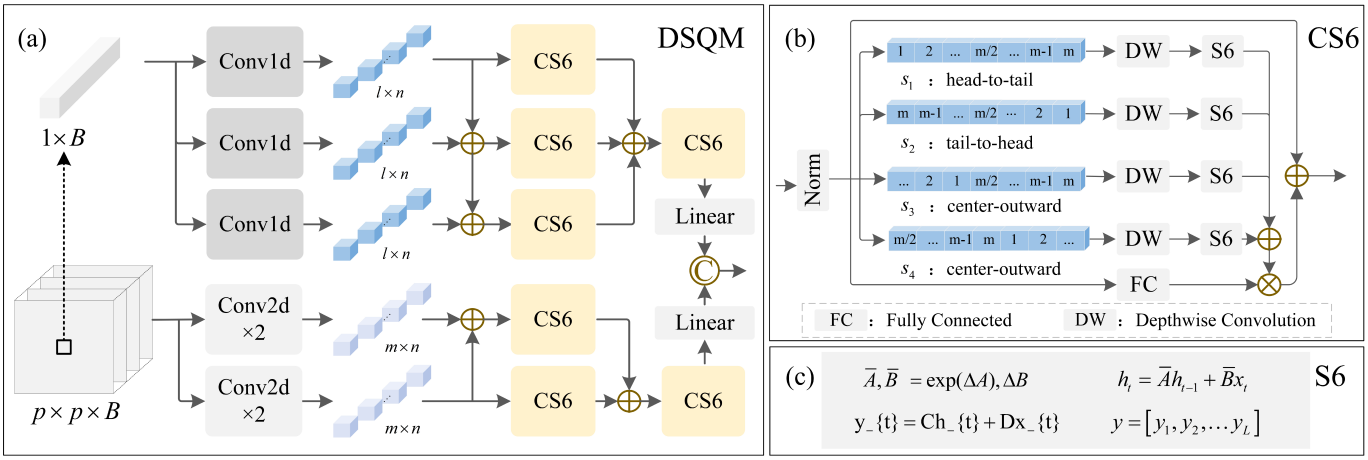


Fig. 3. Flowchart of the feature extraction network DSQM. (a) Overall architecture. (b) Structure of compositing selective SSM (CS6). (c) Computational formula of S6.

per instance is sufficient. Thus, we propose a novel framework where each instance is augmented with a strongly and a weakly perturbed sample, achieving an optimal balance between performance and computational efficiency.

B. Dual-Stream Quad-Scan Mamba

To obtain abundant discriminative spatial–spectral information, a lightweight DSQM feature extraction network is proposed. The network architecture is illustrated in Fig. 3(a).

1) Compositing Selective SSM: S6 captures long-range dependencies with linear complexity, a key advantage for handling high-dimensional data. As shown in Fig. 3(b), the CS6 module is carefully crafted to strengthen contextual modeling by scanning the input sequence from four distinct directions: head-to-tail (s_1), tail-to-head (s_2), and two symmetric center-outward paths (s_3 and s_4). This direction-aware approach allows for comprehensive local–global representation learning in both spectral and spatial dimensions. Building on the S6 framework, CS6 refines the architecture at the structural level, optimizing it for the unique characteristics of HSIs, thereby providing a task-driven enhancement for improved spatial–spectral modeling.

Given the input sequence $\mathbf{F}_{p-1}$, RMSNorm is applied to stabilize training and improve efficiency by avoiding the mean computation. The normalized sequence is then passed through a fully connected layer (FC), producing two distinct representations: $\mathbf{G}_1$ for feature extraction and $\mathbf{G}_2$ for gated operation. The feature branch scans the sequence in four directions: s_1–s_4, with each direction's sequence processed by a 1-D depthwise convolution (DWConv) followed by SiLU activation to capture localized patterns. The processed sequences $\overline{\mathbf{G}}_{1j}$ (for $j \in \{1,2,3,4\}$) are then further refined using the S6 module to capture global dependencies. Finally, the outputs from the four scanned directions are aggregated. The gated block consists of an FC layer followed by the SiLU activation, which generates learnable weights to dynamically filter input features while retaining task-relevant information and suppressing redundancy. The two branches are fused through element-wise multiplication

$$\mathbf{G} = \overline{\mathbf{G}}_1 \odot \mathrm{SiLU}(\mathbf{G}_2). \tag{8}$$

Algorithm 1 Compositing Selective SSM (CS6)

Require: Input feature sequence $\mathbf{F}_{p-1}$: (M, T, K)
Ensure: Output feature sequence $\mathbf{F}_p$: (M, T, K)
/* Normalization and projection */
$\overline{\mathbf{F}}^q_{p-1}$: $(M,T,K) \leftarrow \mathrm{RMSNorm}(\mathbf{F}^q_{p-1})$
$\mathbf{G}_1$: $(M,T,2K) \leftarrow \mathrm{Linear}(\overline{\mathbf{F}}_{p-1})$
$\mathbf{G}_2$: $(M,T,2K) \leftarrow \mathrm{Linear}(\overline{\mathbf{F}}_{p-1})$
/* Feature scanning and processing */
for $j = 1$ to 4 **do**
  $\mathbf{G}_{1j}$: $(M,T,K) \leftarrow s_j(\mathbf{G}_1)$
  /* S6 Processing */
  $\mathbf{B}$: $(M,T,D) \leftarrow \mathrm{Linear}_D(\mathbf{G}_{1j})$
  $\mathbf{C}$: $(M,T,D) \leftarrow \mathrm{Linear}_D(\mathbf{G}_{1j})$
  $\Delta \leftarrow \log(1 + \exp(\mathrm{Broadcast}_K(\mathrm{Linear}_1(\mathbf{G}_{1j})) + \mathrm{Parameter}_\Delta))$
  $\overline{\mathbf{A}}$: $(M,T,K,D) \leftarrow \Delta \odot \mathbf{A}$
  $\overline{\mathbf{B}}$: $(M,T,K,D) \leftarrow \Delta \odot \mathbf{B}$
  $\overline{\mathbf{G}}_{1j}$: $(M,T,K) \leftarrow \mathrm{SSM}(\overline{\mathbf{A}}, \overline{\mathbf{B}}, \mathbf{C})(\mathbf{G}_{1j})$
end for
/* Feature aggregation */
$\overline{\mathbf{G}}_1$: $(M,T,K) \leftarrow \sum_{j=1}^{4} \overline{\mathbf{G}}_{1j}$
/* Gated block */
$\mathbf{G}$: $(M,T,K) \leftarrow \overline{\mathbf{G}}_1 \odot \sigma_2(\mathbf{G}_2)$
/* Residual connection */
$\mathbf{F}_p$: $(M,T,K) \leftarrow \mathbf{F}_{p-1} + \mathrm{Linear}(\mathbf{G})$
return $\mathbf{F}_p$

Finally, a residual connection is applied to preserve information flow, updating the feature as

$$\mathbf{F}_i = \mathbf{F}_{i-1} + \mathrm{Linear}(\mathbf{G}). \tag{9}$$

The proposed CS6 is summarized in Algorithm 1.

2) Pixel Stream: Given the input pixel spectrum $\mathbf{v} \in \mathbb{R}^{1\times B}$, the pixel stream employs 1-D dilated convolutions at different dilation rates to generate multiscale spectral embedding tokens. The process is defined as

$$\mathbf{F}^a_{i-1} = \sigma(\mathrm{Conv1d}(v, k, s, A, d)), \quad a \in \{1,2,3\} \tag{10}$$

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where σ denotes the Leaky ReLU function; k is the kernel size; s is the stride; and d is the dilation rate, set to 1, 2, and 3 to capture shallow spectral information at multiple scales. The number of kernels n is a hyperparameter for feature extraction.

After processing, the pixel vectors are transformed into embedded spectral tokens $\mathbf{F}_{i-1}^{a} \in \mathbb{R}^{l\times n}$, where $l = \lfloor (B-k)/s \rfloor + 1$ is the number of spectral groups. These tokens are fed into a dimensionally matched CS6 module to capture long-range dependencies with linear complexity. The resulting features are fused and enhanced via summation, averaging, and a CS6 module, followed by a linear layer to produce the final multiscale spectral representation

$$\begin{aligned} \mathbf{F}_{i-1}^{a} &= \mathrm{CS6}\left(\mathbf{F}_{i-1}^{a}\right), \quad a \in \{1,2,3\} \\ \mathbf{F}_{\mathrm{pixel}} &= \mathrm{Linear}\left(\mathrm{CS6}\left(\sum_{a=1}^{3} \mathbf{F}_{i-1}^{a}\right)\right), \quad \mathbf{F}_{\mathrm{pixel}} \in \mathbb{R}^{1\times d_0} \end{aligned} \tag{11}$$

where $\mathbf{F}_{\mathrm{pixel}}$ is the final spectral feature, with d_0 set to 64.

3) Patch Stream: Given the input patch $\mathbf{X} \in \mathbb{R}^{p\times p\times B}$, the patch stream uses two parallel 2-D convolution branches to extract shallow spatial features at different scales. Each branch consists of two consecutive convolutions defined as

$$\mathbf{F}_{i-1}^{j} = \mathrm{Conv2D}_2^{j}\left(\mathrm{Conv2D}_1^{j}(\mathbf{X})\right), \quad j = 1,2 \tag{12}$$

where j denotes the branch index, with kernel sizes k_{1j} and k_{2j} and stride sizes s_{1j} and s_{2j} for each convolution. The outputs of both branches are flattened to generate spatial embedding tokens $\mathbf{T}_{i-1}^{j}, j = 1,2$, which are then processed by the CS6 module to capture multiscale discriminative features

$$\mathbf{T}_{i}^{(j)} = \mathrm{CS6}\left(\mathbf{T}_{i-1}^{j}\right), \quad j = 1,2. \tag{13}$$

These spatial features are fused by element-wise averaging, followed by refinement via a CS6 module. The final spatial feature representation is obtained by a linear projection

$$\mathbf{F}_{\mathrm{patch}} = \mathrm{Linear}\left(\mathrm{CS6}\left(\sum_{j=1}^{2} \mathbf{T}_{i-1}^{j}\right)\right), \quad \mathbf{F}_{\mathrm{patch}} \in \mathbb{R}^{1\times d_1} \tag{14}$$

where $\mathbf{F}_{\mathrm{patch}}$ is the final spatial feature, with d_1 set to 16.

The spectral and spatial feature embeddings are concatenated directly. A contrastive head with two linear layers is used to obtain the final representation vector p. In this article, DSQM is utilized for feature extraction in both CL and the autoencoder, with weight sharing during training. Additionally, a lightweight decoder is designed to reconstruct the input patch. The decoder first applies the CS6 module, followed by a transpose convolution for learnable upsampling and a 2-D convolution for feature refinement.

C. Reconstruction Weight-Driven Adaptive Temperature

In general CL frameworks, HPs are often overlooked. Incorrectly pushing HP apart compromises intraclass compactness and increases target–background confusion. The temperature parameter critically controls the optimization of sample pairs. Motivated by this, reconstruction errors are converted into scaling weights that dynamically adjust the temperature parameters of sample pairs.

1) Balanced Ternary Contrastive Loss and Temperature Parameter Analysis (BTCL): In our ternary CL framework, strongly augmented samples $\mathbf{X}_s$ and weakly augmented samples $\mathbf{X}_w$ form positive pairs with the original samples $\mathbf{X}_o$, and negative pairs with augmented samples from other originals. Given a batch size D, for sample $\mathbf{X}_{oi}$, there are 2-D pairs: two positive pairs $(\mathbf{X}_{oi}, \mathbf{X}_{si})$ and $(\mathbf{X}_{oi}, \mathbf{X}_{wi})$, and the rest are negative. BTCL can be computed as follows:

$$\begin{aligned} L_{\mathrm{BTCL}} = \alpha \cdot \left(-\log \frac{\exp\left(\mathrm{sim}\left(\mathbf{p}_{oi}, \mathbf{p}_{si}\right)/\tau\right)}{\sum_{j=1}^{D} \exp\left(\mathrm{sim}\left(\mathbf{p}_{oi}, \mathbf{p}_{sj}\right)/\tau\right)}\right) \\ + \beta \cdot \left(-\log \frac{\exp\left(\mathrm{sim}\left(\mathbf{p}_{oi}, \mathbf{p}_{\mathrm{wi}}\right)/\tau\right)}{\sum_{j=1}^{D} \exp\left(\mathrm{sim}\left(\mathbf{p}_{oi}, \mathbf{p}_{wj}\right)/\tau\right)}\right) \end{aligned} \tag{15}$$

where $\mathbf{p}_o$, $\mathbf{p}_s$, and $\mathbf{p}_w$ are the feature vectors extracted by DSQM; $\mathrm{sim}(\mathbf{a}, \mathbf{b})$ denotes the cosine similarity; τ is the temperature parameter; and α and β balance the loss terms.

In the optimization process, for the negative pair $(\mathbf{p}_{oi}, \mathbf{p}_{sj})$, the magnitude of the gradient with respect to the similarity score is influenced by both the cosine similarity and the temperature. Specifically, the partial derivative of the loss with respect to $\mathrm{sim}(\mathbf{p}_{oi}, \mathbf{p}_{sj})$ is given

$$\frac{\partial L_{\mathrm{BTCL}}}{\partial \mathrm{sim}\left(\mathbf{p}_{oi}, \mathbf{p}_{sj}\right)} = \frac{\alpha}{\tau} \cdot \frac{\exp\left(\mathrm{sim}\left(\mathbf{p}_{oi}, \mathbf{p}_{sj}\right)/\tau\right)}{\sum_{k=1}^{D} \exp\left(\mathrm{sim}\left(\mathbf{p}_{oi}, \mathbf{p}_{sk}\right)/\tau\right)}, \quad \text{for } j \neq i. \tag{16}$$

A higher similarity or a lower temperature leads to a larger gradient magnitude. For high similarity HP, increasing the temperature can attenuate their gradient, thereby reducing their contribution to the optimization objective and suppressing separation in the feature space.

2) Reconstruction Error-Derived Weights: To adaptively mine HP, we design a background reconstruction autoencoder that generates scaling weights from reconstruction errors. First, CEM [29] and a density-based algorithm are applied to efficiently obtain reliable background samples for training. During training, prior sample $\mathbf{X}_{\mathrm{prior}}$, original samples $\mathbf{X}_o$, and background samples $\mathbf{X}_{\mathrm{bck}}$ are fed into the autoencoder for reconstruction. The reconstruction loss for background samples is

$$L_{\mathrm{Rec}} = \frac{1}{p^2} \sum_{i=1}^{p^2} \left\| \mathbf{X}_{\mathrm{bck}} - \mathbf{Y}_{\mathrm{bck}} \right\|_2^2 \tag{17}$$

where p^2 is the total number of pixels in the patch and $\mathbf{Y}_{\mathrm{bck}}$ is the reconstructed patch.

Reconstruction errors $\mathbf{e}$ of all samples are calculated via the mean squared error. Notably, the reconstruction error itself bears a fundamental and intrinsic direct correlation with semantic similarity. For any original sample, considering that the autoencoder is trained on background samples, a larger error is indicative of a higher semantic similarity between the sample and the prior. To capture semantic deviation, we compute the offset vector for each original reconstruction error

$$\mathbf{w}_i = \left| \mathbf{e}_{oi} - \mathbf{e}_{\mathrm{prior}} \right|. \tag{18}$$

Specifically, a smaller offset $\mathbf{w}_i$ signifies a minor semantic deviation from the prior target; in contrast, a larger one implies a more substantial semantic deviation. The scaling weight

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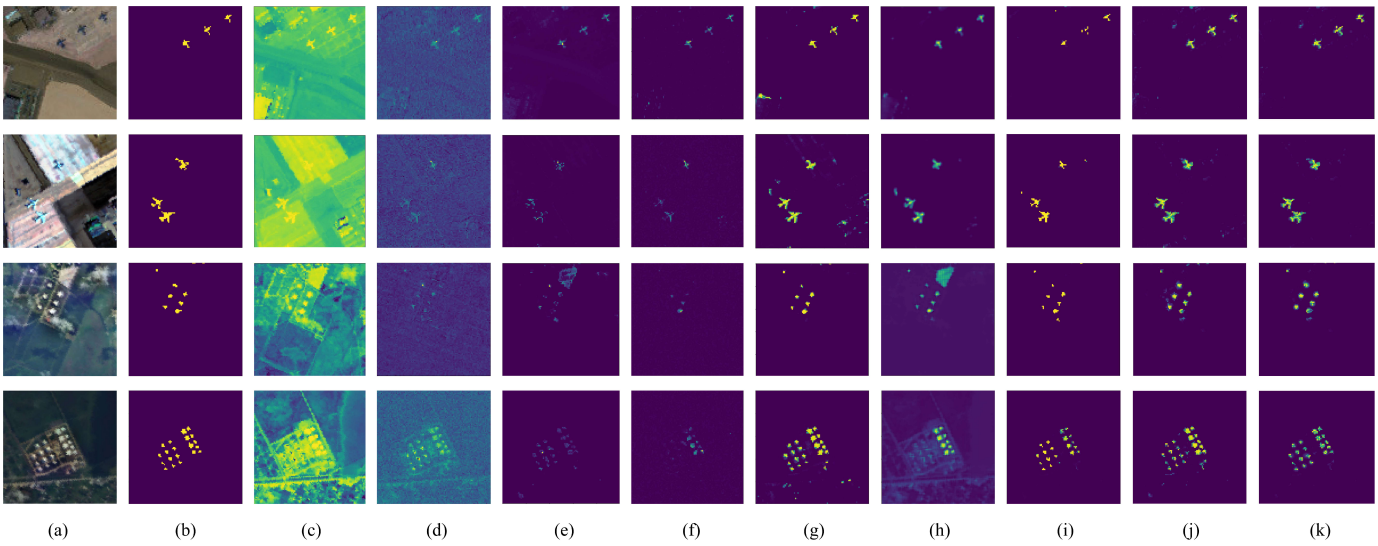


Fig. 4. Experimental datasets and visualized detection maps of competing methods. From top to bottom: Sandiego I, Sandiego II, abu-urban-1, and abu-urban-2. (a) False-color image. (b) Ground truth. (c) SAM. (d) hCEM. (e) CTTD. (f) BLTSC. (g) SSDCTD. (h) HTD-IRN. (i) TSTTD. (j) HTD-Mamba. (k) ATTCA (Ours).

matrix $\mathbf{W} \in \mathbb{R}^{N\times N}$ is built by pairwise Euclidean distances. Samples from target or background groups typically have similar offsets, forming HP with smaller weights, enabling automatic mining and adaptive temperature scaling.

3) Adaptive Temperature Strategy: The scaling weight matrix $\mathbf{W}$ can assign a higher temperature to HP during training. Specifically, the temperature for each negative pair (i, j) is dynamically adjusted as

$$\tau_{ij} = \tau_0 \cdot \exp\left(-w_{ij}\right). \tag{19}$$

Small weights lead to higher temperatures, reducing the focus on optimizing HP, improving intraclass cohesion, and enhancing target feature alignment.

4) Training Loss and Testing Stage: The network's training loss is

$$L = (1-\lambda)\cdot L_{\mathrm{BTCL}} + \lambda \cdot L_{\mathrm{Rec}}. \tag{20}$$

During testing, both prior and test samples are forwarded through the model to extract discriminative features. The cosine similarity is then computed, and a nonlinear transformation is applied to generate the detection map.

IV. Experiments

A. Dataset Description

To evaluate the proposed algorithm, experiments are conducted on four public hyperspectral datasets. Since none provides the prior spectrum, we use the pixel with the highest cosine similarity to all target pixels as the prior.

1) Sandiego Dataset: The Sandiego dataset, collected by AVIRIS over the Sandiego Airport area, California, contains 400 × 400 pixels with a spatial resolution of 3.5 m and spectral resolution of 10 nm. After removing low SNR and water absorption bands, 189 spectral bands are used for target detection. Two regions, Sandiego I (100 × 100 pixels) and Sandiego II (100 × 100 pixels), are selected, with 58 and 134 target pixels (airplanes), respectively. Pseudo-color images with ground-truth annotations are shown in Fig. 4(a) and (b).

2) Urban Dataset: The Urban dataset includes abu-urban-1 and abu-urban-2, collected by AVIRIS over the Texas coast. Both datasets contain 100 × 100 pixels with a spatial resolution of 17.2 m. After removing low SNR and water absorption bands, they retain 204 and 207 bands, respectively, for HTD. The storage tanks, targets in both scenes, occupy 67 and 155 pixels, respectively. Pseudo-color images and ground-truth labels are shown in Fig. 4(a) and (b).

B. Evaluation Metrics

We evaluate the detection performance using visualized detection maps, 3-D receiver operating characteristic (ROC) analysis [30], and target–background separation diagrams.

The detection map is shown as a heatmap, where pixel color and brightness correlate with the likelihood of being a target. The 3-D ROC function is defined by threshold τ, detection probability P_d, and false alarm rate P_f, which can be computed as follows:

$$P_d = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}, \quad P_f = \frac{\mathrm{FP}}{\mathrm{FP}+\mathrm{TN}} \tag{21}$$

where TP, FN, FP, and TN are counts for true positives, false negatives, false positives, and true negatives, respectively. Three 2-D ROC curves, (P_f, P_d), (τ, P_f), and (τ, P_d), assess detection capability, background suppression, and target distinguishability. Ideally, the (P_f, P_d) curve should approach the upper left corner, while (τ, P_f) and (τ, P_d) should be near the lower left and upper right corners, respectively.

For quantification, the area under the curve (AUC) for ROC is computed: $\mathrm{AUC}(P_f, P_d)$, $\mathrm{AUC}(\tau, P_f)$, and $\mathrm{AUC}(\tau, P_d)$, with higher values (1 being optimal) indicating better performance. A lower $\mathrm{AUC}(\tau, P_f)$ is preferred, with 0 being ideal. Additionally, two composite metrics [31] are used

$$\mathrm{AUC}_{\mathrm{OA}} = \mathrm{AUC}_{(P_f,P_d)} + \mathrm{AUC}_{(\tau,P_d)} - \mathrm{AUC}_{(\tau,P_f)} \tag{22}$$

$$\mathrm{AUC}_{\mathrm{SNPR}} = \frac{\mathrm{AUC}_{(\tau,P_d)}}{\mathrm{AUC}_{(\tau,P_f)}} \tag{23}$$

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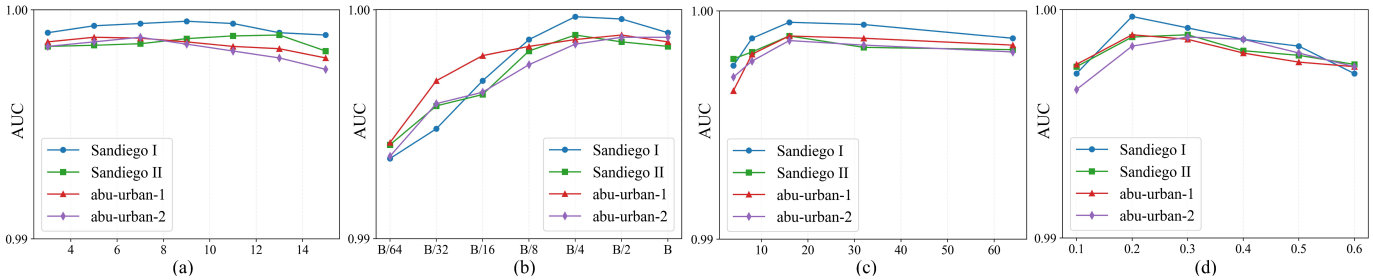


Fig. 5. Parameter impact analysis experiments on four datasets, in which the peak represents the highest accuracy. (a) Patch size. (b) Quantization bins. (c) Token embedding dimension. (d) Weighting coefficient of the loss.

where higher values of AUC_{OA} and AUC_{SNPR} reflect better performance (ideal values: 2 and ∞).

Finally, the target–background separation diagram, based on Box–Whisker plots, visualizes separability. The magenta and blue boxes represent target and background pixels, respectively. A greater distance between the boxes indicates better separability, with higher magenta boxes and lower blue boxes showing better detection and background suppression.

C. Implementation Details

In the proposed ATTCA, the input patch sizes p are set to 9, 13, 5, and 7 for the four datasets. During data augmentation, the number of quantized channels λ is $B/2$, where B represents the spectral dimension. For feature extraction, the quantization intervals N are configured as $B/4$, $B/4$, $B/2$, and $B/2$ for the four datasets. Feature dimensions for the spectral and spatial streams are 64 and 16, respectively. The token counts (m_1, m_2) are set to (30, 25), (15, 25), (20, 15), and (15, 15) per dataset, and the embedding dimensions n are 16 for spatial and spectral streams. The temperature parameter τ is 0.1, with contrastive loss parameters α and β set to 0.2 and 0.8, respectively. The reconstruction loss coefficients λ are 0.2, 0.3, 0.2, and 0.3 for the datasets. Furthermore, the batch size and epoch are set as 120 and 100 during the training stage, respectively. The AdamW optimizer is used with weight decay 10^{-4}, and the learning rate starts at 10^{-4}, increasing with a linear warm-up for ten epochs before decaying via a cosine annealing scheduler. ATTCA is implemented in Python 3.8.19 with PyTorch 1.13.0+cu117, running on an Intel Core i9-10900X CPU, 32 GB of RAM, and an NVIDIA GeForce RTX 2080 Ti with 11 GB of GPU memory.

D. Parameter Selection

1) Patch Size p: To retain spatial information, the data are divided into patches. Patch size significantly affects augmented data quality and feature extraction. We experimented with p ranging from 5 to 15 in increments of 2 across four datasets. The results show that different datasets exhibit varying sensitivities to patch size, influenced by the size of target objects, which affects the detection performance. As shown in Fig. 5(a), we set p to 9, 11, 5, and 7 for the four datasets to optimize spatial information while minimizing noise.

2) Quantization Intervals N: The quantization intervals N in RQ control the noise magnitude introduced to the model. We experimented with N: B, B/2, B/4, B/8, B/16, B/32, and B/64, where B is the number of spectral channels. The impact of N on detection performance is shown in Fig. 5(b). For balanced training, we set N to B/4, B/2, B/2, and B/2.

3) Token Embedding Dimension n: We use spatial and spectral embedding modules to process patch and pixel data, generating spatial and spectral tokens. To align and fuse these representations, both types of token have the same embedding dimension n. We experimented with n values from 8, 16, 32, 64, 128, and 256 across four datasets, as shown in Fig. 5(c). Detection performance improves with n up to a point, after which it stabilizes or slightly declines due to redundancy in high-dimensional features. To balance accuracy and efficiency, we set $n = 16$ for all datasets.

4) Weighting Coefficient of the Loss λ: λ controls the contributions of reconstruction and triplet CL loss. We experimented with λ values from 0.1 to 0.6 in steps of 0.1. As shown in Fig. 5(d), accuracy increases with λ until reaching an optimal point, after which it declines. Small λ values maintain a balance between feature alignment and interclass discrimination, while large λ values prioritize target alignment, reducing target–background separability. The optimal λ values for the four datasets are set to 0.2, 0.3, 0.2, and 0.3.

E. Comparison Methods

We compare our method against eight representative and publicly available HTD approaches: spectral angle mapper, hierarchical constrained energy minimization (hCEM) [32], chessboard-shaped topology for HTD (CTTD)[1] [33], background learning based on target suppression constraint (BLTSC)[2] [34], self-supervised deep clustering-based target detector (SSDCTD)[3] [35], HTD based on an interpretable representation (HTD-IRN)[4] [36], triplet spectral-wise Transformer-based target detector (TSTTD)[5] [37], and HTD with pyramid SSM (HTD-Mamba)[6] [18]. Official codes are used for all methods whenever available.

To ensure fairness and reproducibility, we adopt the recommended hyperparameter settings from the original articles on their respective datasets. For datasets not covered in the original works, we perform fine-grained parameter tuning.

[1]https://github.com/sxt1996/CTTD
[2]https://github.com/zhangxin-xd/BLTSC
[3]https://github.com/zxd52csx/Pixel-level-Hyperspectral-Target-Detection
[4]https://github.com/shendb2022/HTD-IRN
[5]https://github.com/shendb2022/TSTTD
[6]https://github.com/shendb2022/HTD-Mamba

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Specifically, for hCEM, the two key parameters λ and ϵ are set to (200, 0.004), (200, 0.002), (180, 0.006), and (180, 0.002), respectively. For CTTD, we set the intervals for vertical and horizontal division (ver, hor) as (18, 2), (16, 3), (19, 4), and (19, 6), respectively. In BLTSC, the threshold k used for coarse detection is set to 0.15. Meanwhile, the parameter λ is assigned values of 10^3, 10^4, 10^3, and 10^3. For SSDCTD, the key parameter, the weight of implicit CL w is set to 0.52, 0.5, 0.38, and 0.32. For HTD-IRN, the background endmembers m and the regularization parameter of sparse loss η_1 are set to (30, 10), (30, 0.1), (15, 0.01), and (15, 0.1). For the TSTTD, spectral group length is set to 30, 20, 40, and 30 for four datasets. Furthermore, for the spectral Transformer module, the head dimension, hidden dimension, and depth are set to (64, 64, 4), (128, 128, 2), (64, 64, 4), and (64, 64, 4). For the HTD-Mamba, we uniformly set the embedding size N and network depth l to 16 and 1. In addition, the input patch size p and spectral group length m are set as follows: (11, 30), (13, 5), (5, 5), and (11, 5).

F. Experimental Results

To comprehensively evaluate the proposed ATTCA method for HTD tasks, we provide visualized detection maps, ROC curve analyses, and target–background separation diagrams across four benchmark datasets. Fig. 4 presents the detection maps of eight comparison methods and ATTCA. ATTCA strikes a good balance between target enhancement and background suppression. In the Sandiego II dataset with three aircraft as targets, SAM and hCEM delineate complete target contours but produce many false alarms. CTTD and BLTSC effectively suppress the background but struggle to preserve the target shapes. HTD-IRN and TSTTD show lower false alarm rates but miss parts of the targets. SSDCTD and HTD-Mamba perform well in target localization but miss some details, with higher false alarm rates. Compared to HTD-Mamba, ATTCA achieves superior alignment of target characteristics, which can be evidenced by the upper aircraft in the detection map.

In the abu-urban-1 dataset, where six small storage tanks are the targets, SAM, hCEM, CTTD, and HTD-IRN suffer from significant false alarms. BLTSC misses several target regions, while TSTTD and SSDCTD perform well in highlighting targets but miss some small ones. SSDCTD's self-supervised clustering mechanism emphasizes spectral distinctiveness, suppressing background but leading to missed targets. Both HTD-Mamba and ATTCA restore target contours, but ATTCA better suppresses false alarms in complex backgrounds.

Fig. 6 shows the 3-D and 2-D ROC curves. ATTCA consistently achieves the top-left position in the (P_f, P_d) curves, indicating superior performance. It also ranks among the best in the (τ, P_f) curve and the highest in (τ, P_d), demonstrating robust target enhancement and background suppression. Compared with the competitive HTD-Mamba, the curve (τ, P_d) of ATTCA lies closer to the upper left corner, due to its RWAT strategy that effectively mines HP samples and aligns target features. Meanwhile, its curve (τ, P_f) shifts toward the lower left corner, benefiting from the ternary CL framework that balances robustness and representation consistency, thus enhancing generalization and suppressing false alarms. Additionally, the target–background separation diagrams in Fig. 7 show that ATTCA achieves larger separation with minimal overlap, demonstrating superior discriminability. Although SSDCTD, TSTTD, and HTD-Mamba also suppress backgrounds, their target boxes overlap more, whereas ATTCA ensures better separation and higher target localization.

As shown in Table I, ATTCA achieved top-two rankings in all five evaluation metrics except for ranking third in AUC(τ, P_f) on the abu-urban-1 dataset. Comprehensive evaluation results demonstrate that ATTCA enhances target responses while effectively suppressing background interference, achieving a robust balance between sensitivity and specificity. Its consistent performance across complex scenes further verifies its strong generalization and robustness.

G. Ablation Studies

1) Effectiveness of S-WFI: In S-WFI, each original sample is paired with strong and weak augmentation to form positive pairs. The augmentation ratio is determined through experiments. As shown in Fig. 8, we evaluated eight combinations of strong (S) and weak (W) augmentations. The heatmaps show that the (1, 1) configuration achieves the highest detection accuracy across all datasets, with an average gain of 0.29%, and slightly outperforms (2, 2) by 0.04%. Excessive augmentation pairs tend to slow the convergence of contrastive loss and destabilize optimization, leading to reduced accuracy. Under a fixed augmentation budget, a higher proportion of weak augmentations yields about 0.55% higher accuracy, suggesting that overly strong perturbations may disrupt semantic consistency and hinder feature alignment. Therefore, the configuration (1, 1) is adopted.

2) Effectiveness of DSQM: We conducted ablation experiments, evaluating the contributions of the pixel branch, patch branch, and four-directional scanning operation in the CS6 module. The results in Table II show that removing either the patch or pixel branch reduces partial spatial or spectral information, degrading detection accuracy. In particular, removing the patch branch yields an average accuracy reduction of 0.06%, whereas removing the pixel branch results in a more pronounced 0.24% drop, demonstrating the complementary nature of the dual-stream architecture and the critical importance of the pixel branch. Furthermore, the absence of four-directional scanning operation in the CS6 module leads to a performance decline of 0.1%, demonstrating its indispensable role in the capture of hierarchical local–global dependencies.

3) Effectiveness of RWAT: To address the impact of HP in CL loss optimization, we introduce an adaptive temperature strategy based on reconstruction error-weighted adjustment. This method uses the reconstruction error of the background autoencoder to dynamically adjust the temperature parameter in CL. As shown in Table II, compared to a fixed temperature of 0.1, RWAT allows the network to identify hidden positive samples and adjust the temperature for better feature alignment. This strategy mitigates HP, reduces intraclass distances, and improves the detection accuracy by 0.05% on average.

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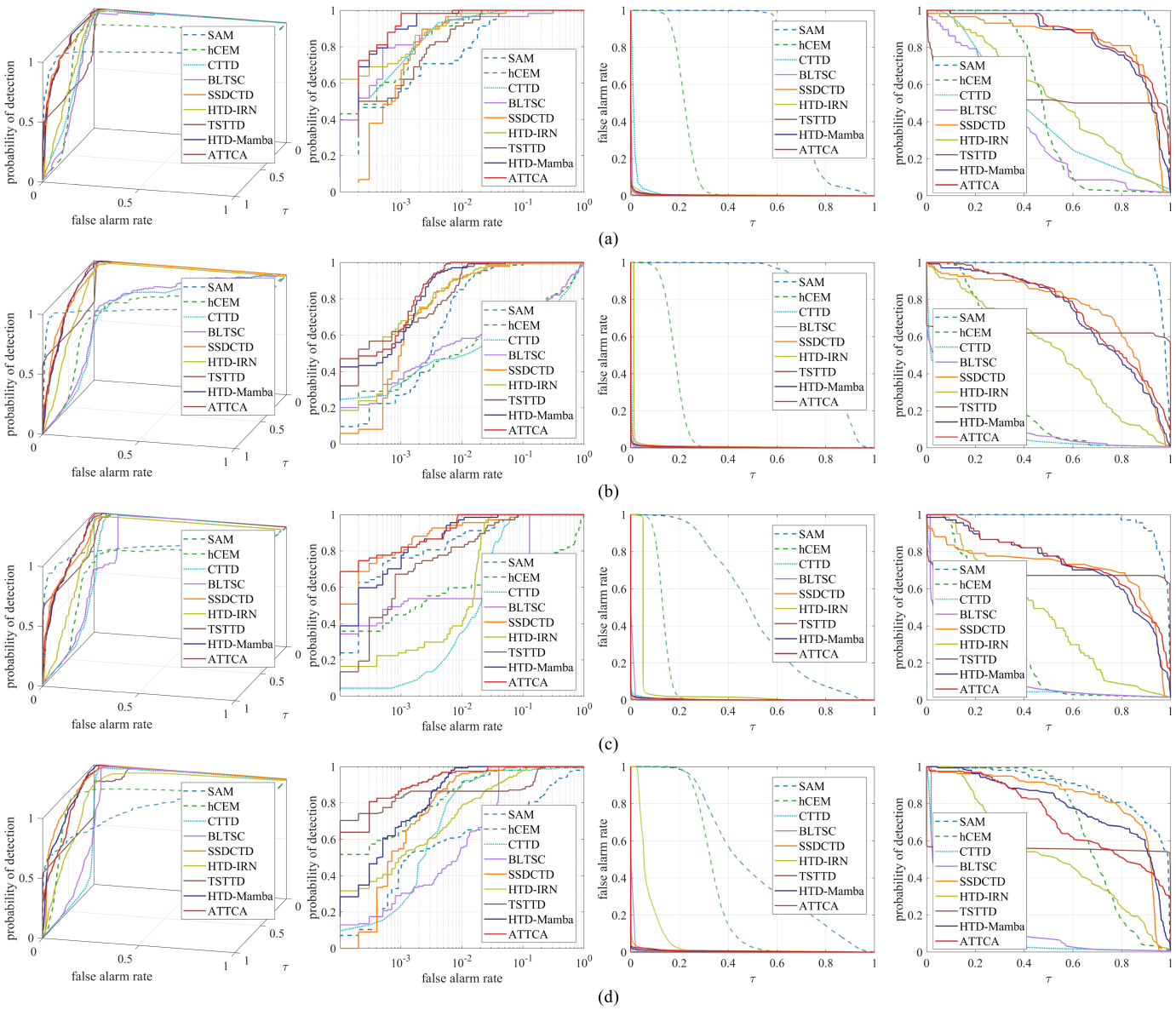


Fig. 6. ROC curves for each detection method on the four datasets, from left to right: 3-D ROC curves, 2-D ROC curves (P_f and P_d), 2-D ROC curves (τ and P_d), and 2-D ROC curves (τ and P_f). (a) Sandiego I. (b) Sandiego II. (c) Abu-urban-1. (d) Abu-urban-2.

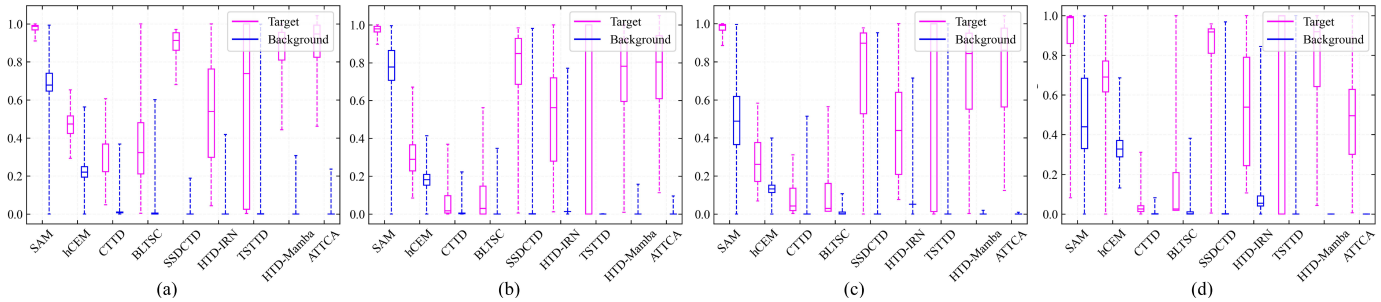


Fig. 7. Target–background separation diagrams of competing methods on four datasets. (a) Sandiego I. (b) Sandiego II. (c) Abu-urban-1. (d) Abu-urban-2.

H. Discussion of Testing Time and Calculation Cost

Computational cost and detection time are key metrics for evaluating detector performance. We report the detection time of various detectors on four datasets to assess runtime efficiency. For DL-based approaches, parameter counts and FLOPs are also measured to analyze computational complexity, as summarized in Table III. Traditional data-driven methods such as SAM, hCEM, and the topological method CTTD

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TABLE I

QUANTITATIVE RESULTS OF DIFFERENT METHODS FOR FOUR DATASETS. BOLD REPRESENTS THE BEST RESULT AND UNDERLINED REPRESENTS THE SECOND

Dataset	Metric	SAM	hCEM	CTTD	BLTSC	SSDCTD	HTD-IRN	TSTTD	HTD-Mamba	ATTCA
Sandiego I	$AUC_{(P_f,P_d)} \uparrow$	0.9938	0.9976	0.9981	0.9917	0.9979	0.9987	0.9973	**0.9996**	0.9995
	$AUC_{(\tau,P_f)} \downarrow$	0.6969	0.2221	0.0055	0.0033	0.0042	0.0036	**0.0014**	0.0039	0.0031
	$AUC_{(\tau,P_d)} \uparrow$	**0.9773**	0.4849	0.3461	0.3587	0.8343	0.5328	0.5271	0.8481	0.8755
	$AUC_{OA} \uparrow$	1.2742	1.2604	1.3388	1.3472	1.8280	1.5279	1.5229	1.8438	**1.8700**
	$AUC_{SNPR} \uparrow$	1.40	2.18	62.93	109.50	198.64	148.00	**366.01**	217.68	282.42
Sandiego II	$AUC_{(P_f,P_d)} \uparrow$	0.9932	0.9599	0.9959	0.9499	0.9976	0.9971	0.9973	0.9985	**0.9989**
	$AUC_{(\tau,P_f)} \downarrow$	0.7821	0.1672	0.0286	0.0058	0.0081	0.0035	**0.0010**	0.0040	0.0033
	$AUC_{(\tau,P_d)} \uparrow$	**0.9752**	0.5156	0.2263	0.1424	0.7490	0.5484	0.6237	0.7247	0.7503
	$AUC_{OA} \uparrow$	1.1862	1.3083	1.1197	1.0864	1.7368	1.5420	1.6200	1.7192	**1.7460**
	$AUC_{SNPR} \uparrow$	1.25	3.08	7.91	24.48	92.47	156.69	**643.01**	181.85	227.36
abu-urban-1	$AUC_{(P_f,P_d)} \uparrow$	0.9944	0.8937	0.9180	0.9461	0.9968	0.9884	0.9935	0.9983	**0.9989**
	$AUC_{(\tau,P_f)} \downarrow$	0.5016	0.5893	0.3192	0.0051	**0.0009**	0.0603	0.0015	0.0032	0.0029
	$AUC_{(\tau,P_d)} \uparrow$	**0.9719**	0.7842	0.6865	0.1311	0.7047	0.4587	0.6783	0.7307	0.7558
	$AUC_{OA} \uparrow$	1.4648	1.0886	1.2853	1.0721	1.7006	1.3868	1.6703	1.7248	**1.7519**
	$AUC_{SNPR} \uparrow$	1.94	1.33	2.15	25.63	**783.00**	7.61	464.58	173.98	260.62
abu-urban-2	$AUC_{(P_f,P_d)} \uparrow$	0.9112	0.9239	0.9049	0.9846	0.9969	0.9881	0.9965	0.9986	**0.9988**
	$AUC_{(\tau,P_f)} \downarrow$	0.5060	0.1457	0.0233	0.0060	0.0087	0.0719	0.0043	0.0041	**0.0037**
	$AUC_{(\tau,P_d)} \uparrow$	**0.8896**	0.4331	0.1655	0.1371	0.8195	0.5250	0.5968	0.7862	0.8357
	$AUC_{OA} \uparrow$	1.2947	1.2113	1.0471	1.1157	1.8077	1.4412	1.5890	1.7807	**1.8309**
	$AUC_{SNPR} \uparrow$	1.46	2.97	7.10	22.67	94.19	7.30	138.79	189.93	**225.86**

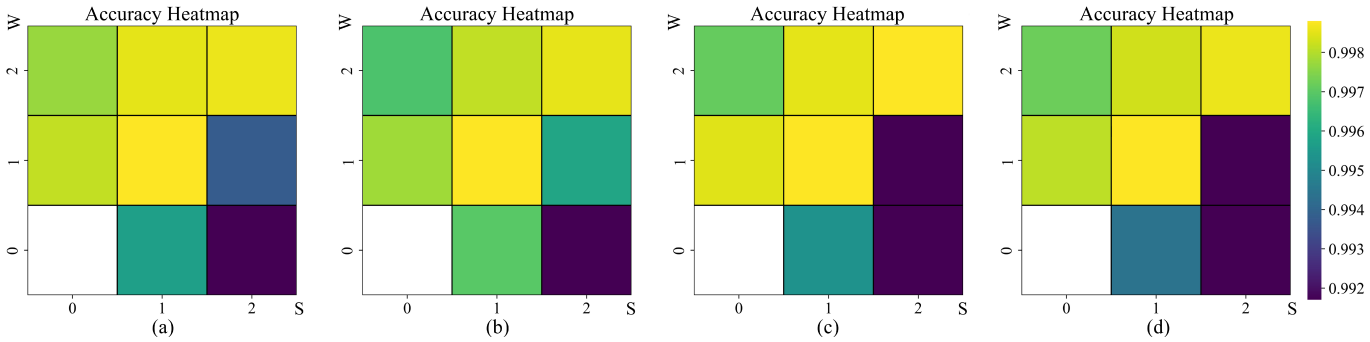


Fig. 8. Detection accuracy heatmap under different combinations of strong (S) and weak (W) augmentations across four datasets. The x-axis and y-axis represent the number of S and W, respectively. Color intensity indicates detection accuracy, with darker regions representing poor performance. (a) Sandiego I. (b) Sandiego II. (c) Abu-urban-1. (d) Abu-urban-2.

TABLE II

EFFECTIVENESS OF INDIVIDUAL MODULES WITHIN THE ATTCA FRAMEWORK, EVALUATED BY AUC (P_f AND P_d) ON FOUR DATASETS. BOLD INDICATES THE BEST RESULT (THE COMPLETE ATTCA STRUCTURE)

methods	Sandiego I	Sandiego II	abu-urban-1	abu-urban-2
ATTCA	**0.9995**	**0.9989**	**0.9989**	**0.9988**
ATTCA without pixel stream	0.9976	0.9954	0.9972	0.9963
ATTCA without patch stream	0.9991	0.9982	0.9984	0.9979
ATTCA without Quad-scan	0.9986	0.9982	0.9979	0.9973
ATTCA without Adaptive Temperature	0.9992	0.9985	0.9983	0.9981

run faster but yield lower accuracy. In contrast, DL-based detectors achieve high accuracy with competitive runtime efficiency. Notably, ATTCA attains superior detection speed and consistently lower FLOPs across all datasets, particularly on Sandiego I and abu-urban-1, while using fewer parameters than most counterparts. Overall, ATTCA achieves an excellent bal-

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TABLE III
DETECTION TIME, PARAMETERS, AND FLOPS OF DIFFERENT DETECTORS ON FOUR DATASETS. BOLD REPRESENTS THE BEST RESULT AND UNDERLINED REPRESENTS THE SECOND

dataset		SAM	hCEM	CTTD	BLTSC	SSDCTD	HTD-IRN	TSTTD	HTD-Mamba	ATTCA(Ours)
Sandiego I	Detection Times(s)↓	0.4448	0.6138	**0.1935**	0.3856	0.4988	4.0984	6.3567	4.2404	2.8017
	Params(M)↓	/	/	/	1.5660	**0.0225**	0.4090	0.8867	0.3252	0.4238
	FLOPS(G)↓	/	/	/	**0.0031**	0.4701	2.2021	0.1523	0.1398	0.0343
Sandiego II	Detection Times(s)↓	0.4419	0.8301	**0.1898**	0.3698	0.5119	4.0823	3.6235	4.3030	3.6385
	Params(M)↓	/	/	/	1.5624	**0.0287**	0.4090	0.9254	0.4907	0.4802
	FLOPS(G)↓	/	/	/	**0.0032**	0.5644	2.2021	0.1695	1.1704	0.0414
abu-urban-1	Detection Times(s)↓	0.4647	0.6341	**0.2013**	0.4469	0.5014	4.1399	2.1423	4.3286	1.3288
	Params(M)↓	/	/	/	1.5780	**0.0233**	0.4092	0.9354	0.3252	0.4197
	FLOPS(G)↓	/	/	/	**0.0032**	0.4782	2.2036	0.1678	0.1398	0.0090
abu-urban-2	Detection Times(s)↓	0.5826	0.6093	**0.1956**	0.3635	0.5062	4.1943	2.6528	3.4741	1.4528
	Params(M)↓	/	/	/	1.5805	**0.0235**	0.4096	0.8361	0.3252	0.7170
	FLOPS(G)↓	/	/	/	**0.0031**	0.4798	2.2077	0.1588	0.1398	0.0300

ance of computational cost, parameter efficiency, and detection speed, making it well-suited for real-time and resource-limited applications.

V. CONCLUSION

In this work, we propose the ATTCA for HTD. First, the S-WFI augmentation technique generates strongly and weakly augmented samples, paired with the original to form a ternary CL framework, balancing model robustness and feature consistency. Next, a DSQM network based on CS6 is proposed to extract multiscale spatial–spectral information, efficiently modeling long-range dependencies with low complexity and fewer parameters. Finally, the RWAT strategy adaptively transfers reconstruction errors to weights and dynamically adjusts the contrastive loss temperature, enhancing HP identification and improving intraclass compactness. Experiments on four benchmark hyperspectral datasets demonstrate ATTCA's superiority in accuracy, robustness, and efficiency. Building on this direction, future work holds great potential for further innovation, especially in incremental learning [38], resource-constrained systems, and real-time remote sensing.

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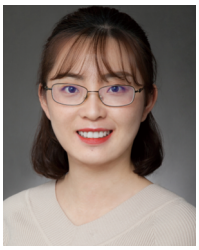
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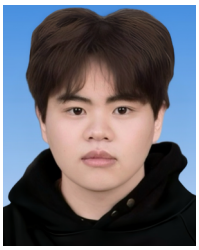
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